05-15 第1篇

阅读理解（说明文） How a Simple Lemon Can Power a Clock

① A school science teacher once walked into the classroom carrying a lemon, two small metal nails and a tiny digital clock. She stuck the nails into the lemon, connected thin wires to the clock — and, to the surprise of every student, the clock began to blink. "No batteries today," she smiled. "Our lemon is the battery." How can a small yellow fruit really power a clock? The answer opens a door to the science of **electricity**.

② Electricity, at its simplest, is a **flow of tiny particles** called electrons. In a common battery, the battery pushes electrons through a wire from its "minus" end to its "plus" end. This movement is what makes a clock show numbers, a torch light up, or a speaker play sound. Without this quiet flow, the electronic devices on our tables are only pieces of plastic and metal.

③ A lemon, surprisingly, also can push electrons — though only a little. The secret is in **lemon juice**, which is a mild **acid**. Acids are liquids that contain free chemical pieces ready to take part in reactions. When the teacher pushes two **different** metals into the lemon, one metal — for example, zinc on the nail — loses electrons more easily. The other metal — often copper on a coin or wire — gains them more easily.

④ So the zinc end slowly sends out electrons. The electrons travel through the wires, through the clock's tiny circuits, and back to the copper end. On the way through the clock, they light up the numbers. The lemon is not losing its fruit power; rather, the metals and the acid are doing most of the chemistry work. The lemon is simply providing a "wet stage" where the two metals can meet the acid in a controlled way.

⑤ One single lemon only produces a small amount of electricity — far too little for a phone or a fridge. But several lemons connected in a row can add their power together, the way a team of small dogs can pull a sled that one dog alone cannot. In one well-known class experiment, students connected five large lemons, plus pairs of zinc-and-copper strips, to power a small LED bulb for a few minutes.

⑥ Real batteries in our phones work on the same basic idea: two different metals and a special liquid or paste between them. Modern batteries are far more powerful, lighter, and safer than any fruit. Yet a lemon battery is still a wonderful science gift. It reminds us that electricity, which feels like a modern mystery, is actually a very old natural trick hiding inside many common things around us — including the quiet yellow fruit in the kitchen fridge.

根据短文内容，选择最佳答案。

1. What did the teacher connect to the clock to make it work?

A. A banana and one metal strip.
B. A lemon and two different metal pieces.
C. Two batteries.
D. Some sugar water and a spoon.

2. What is electricity described as in paragraph ②?

A. A heavy liquid.
B. A flow of tiny particles called electrons.
C. A ball of hot fire.
D. A strong magnet.

3. Why are TWO DIFFERENT metals needed in the lemon experiment?

A. To make the lemon sweeter.
B. Because one metal loses electrons more easily while the other gains them more easily, creating a flow.
C. Because two metals look prettier than one.
D. To scratch the lemon's skin.

4. What role does the lemon play in the experiment?

A. The lemon is the only source of all electricity.
B. The lemon provides a "wet stage" where two metals meet the acid.
C. The lemon cools the metals.
D. The lemon catches the electrons and eats them.

5. Which is the BEST main idea of the passage?

A. Lemons are the best fruit to eat.
B. Electricity is very dangerous and should not be touched.
C. A lemon and two different metals can act as a small battery, and real batteries use the same basic idea.
D. Students should only study chemistry, not physics.

答案与解析

1.
【答案】B
【题型】细节理解题

【定位】第①段第1–2句 "a lemon, two small metal nails..."
【解析】老师用柠檬和两个不同金属片。A/C/D 均不符。故选 B。

2.
【答案】B
【题型】词义理解题
【定位】第②段第1–2句 "Electricity... is a flow of tiny particles called electrons."
【解析】电是电子的流动。A/C/D 均不符。故选 B。

3.
【答案】B
【题型】细节理解/推理判断题
【定位】第③段 "one metal... loses electrons more easily. The other metal... gains them more easily."
【解析】两种不同金属才能形成电子流动。A/C/D 均无据或偏题。故选 B。

4.
【答案】B
【题型】细节理解题
【定位】第④段末句 "The lemon is simply providing a 'wet stage' where the two metals can meet the acid..."
【解析】柠檬提供"湿舞台"让金属与酸接触。A 与原文矛盾，C/D 无据。故选 B。

5.
【答案】C
【题型】主旨大意题
【定位】第⑤⑥段总结
【解析】主旨即"柠檬加两种金属可当小电池，真正电池亦同理"。A/B/D 均不符。故选 C。

词汇注释（超纲词）

- **electron** [n.] 电子
- **zinc** [n.] 锌
- **copper** [n.] 铜
- **LED bulb** [n. phr.] LED 小灯